

1. Apply Apriori Algorithm to find frequent itemsets from the following transaction database. Consider minimum support threshold = 2.

Ans:

Given database is:

&

Support count = 2

TID	List of items
T1	I1, I2, I3
T2	I2, I3, I4
T3	I4, I5
T4	I1, I2, I4
T5	I1, I2, I3, I5
T6	I1, I2, I3, I4

Step 1: i) Create a table containing support count of each item present in dataset - called C1 (candidate set).

Item	Count
I1	4
I2	5
I3	4
I4	4
I5	2

ii) prune step: Compare candidate set item's support count with minimum support count. The above table shows that all items have a support count ≥ 2 , so all are frequent.

$$L_1 = \{I_1, I_2, I_3, I_4, I_5\}$$

Step 2: i) Join step: Generate candidate set C2 (2-item set) using L1. And find out the occurrences of 2-itemset from the given dataset.

Itemset	Support Count	Pruned
$I_1 I_2$	4	
$I_1 I_3$	3	
$I_1 I_4$	2	
$I_1 I_5$	1	pruned ✓
$I_2 I_3$	4	
$I_2 I_4$	3	
$I_2 I_5$	1	pruned ✓
$I_3 I_4$	2	
$I_3 I_5$	1	pruned ✓
$I_4 I_5$	1	pruned ✓

ii) Prune step: After pruning,

$$L_2 = \{ \{I_1, I_2\}, \{I_1, I_3\}, \{I_1, I_4\}, \{I_2, I_3\}, \{I_2, I_4\}, \{I_3, I_4\} \}$$

Step 3: ii) Join step: Generate set C_3 (3-item set).

Itemset	Support Count	Pruned
$I_1 I_2 I_3$	3	
$I_1 I_2 I_4$	2	
$I_1 I_3 I_4$	1	pruned ✓
$I_2 I_3 I_4$	2	

ii) Prune step: After pruning,

$$L_3 = \{ \{I_1, I_2, I_3\}, \{I_1, I_2, I_4\}, \{I_2, I_3, I_4\} \}$$

Step 4: i) Join step: Generate set of C_4 (4-item set).

$$C_3 = \{ \{I_1, I_2, I_3, I_4\} \}$$

ii) Prune step: Its 3-item set subsets are $\{I_1, I_2, I_3\}$, $\{I_1, I_2, I_4\}$, $\{I_2, I_3, I_4\}$, and $\{I_1, I_3, I_4\}$. $\{I_1, I_3, I_4\}$ is not in L_3 .

$$\therefore L_4 = \{ \}$$

All frequent item sets are:

$$L_1 = \{ \{I_1\}, \{I_2\}, \{I_3\}, \{I_4\}, \{I_5\} \}$$

$$L_2 = \{ \{I_1, I_2\}, \{I_1, I_3\}, \{I_1, I_4\}, \{I_2, I_3\}, \{I_2, I_4\}, \{I_3, I_4\} \}$$

$$L_3 = \{ \{I_1, I_2, I_3\}, \{I_1, I_2, I_4\}, \{I_2, I_3, I_4\} \}$$

2.

Apply the FP-Growth Algorithm to find frequent itemsets from the following transaction database. Assume the minimum support count is 2 and minimum confidence threshold = 70%.

Ans:

Given dataset is:

and

Support count is 2.

T-ID	List of items
T1	A, C, D
T2	B, C, E, F
T3	A, B, C, E
T4	B, E
T5	A, C, E

Items	Support count
A	3
B	3
C	4
D	1
E	4
F	1

The items D and F are pruned since they are not minimum support.

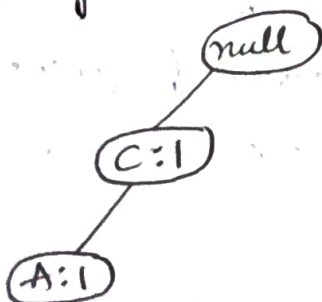
$L = \{C:4, E:4, A:3, B:3\}$

Now, for each transaction the respective ordered item set is built.

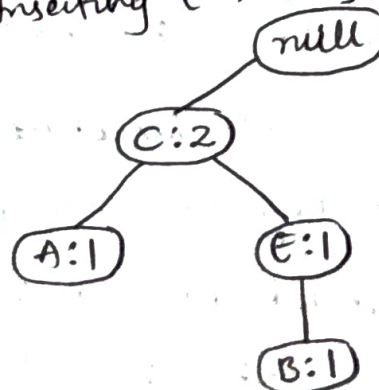
Trans ID	Items	Ordered-Itemsets
T1	{A, C, D}	{C, A}
T2	{B, C, E, F}	{C, E, B}
T3	{A, B, C, E}	{C, E, A, B}
T4	{B, E}	{E, B}
T5	{A, C, E}	{C, E, A}

Construct FP-tree :=

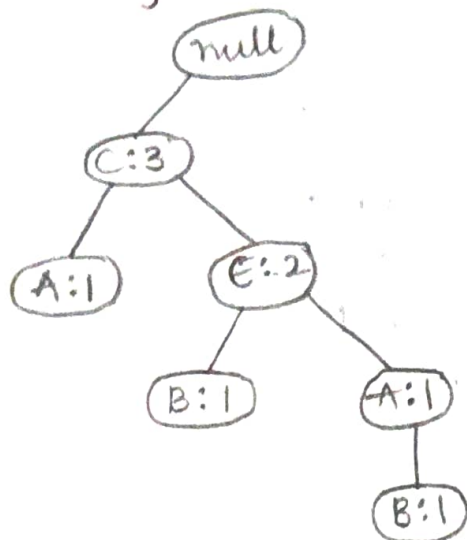
a) Inserting {C, A}



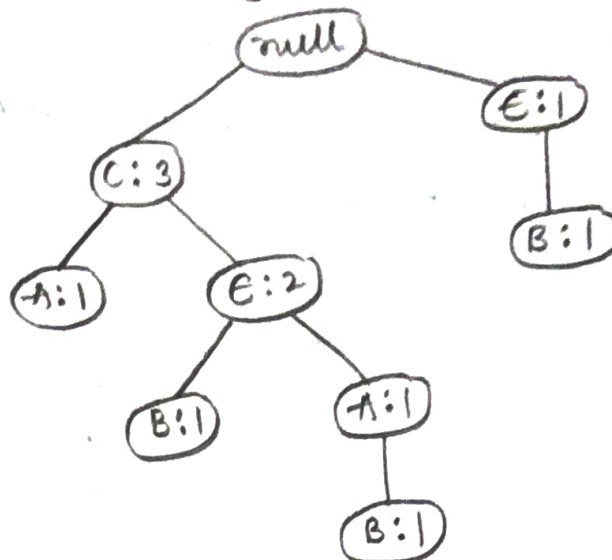
b) Inserting {C, E, B}



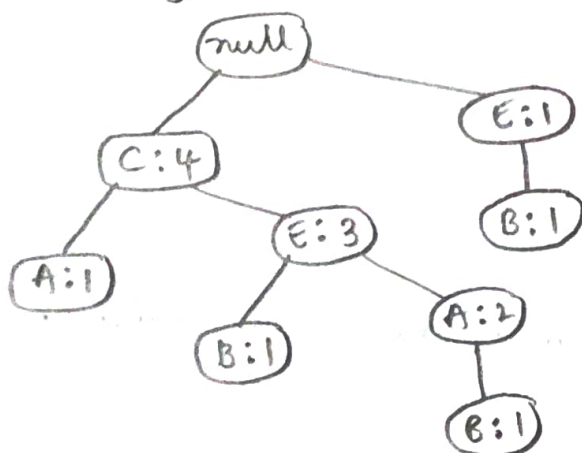
c) Inserting {C, E, A, B}



d) Inserting {E, B}



e) Inserting {C, E, A}



Item	Pattern Base	Conditional FP-tree	Frequent item sets
A	{C:1}, {C,E:2}	{C:3} {E:2}	{A,C}:3, {A,E}:2, {A,C,E}:2
B	{E:1}, {C,E,1}, {C,E,A:1}	{C:2} {E:3}	{B,C}:2, {B,E}:3, {B,C,E}:2
C	{}		
E	{C:3}	{C:3}	{E,C}:3

3.a. Describe the steps involved in constructing the FP-tree, generating frequent itemsets & how it improves efficiency compared to the Apriori Algorithm.

Ans: FP-Tree Construction & Frequent Itemset Generation:

FP-Tree is a data structure used to find frequent itemsets without generating large no. of candidate sets like Apriori algo.

Steps to construct FP-tree:

1) Scan the database: Read all transaction & count the frequency

of each item.

2) Remove infrequent item: Items whose support is below the minimum support threshold are removed.

3) Sort frequent items: Arrange the remaining items in descending order of their frequency.

4) Create the FP-tree: Scan each transaction again & insert its frequent items into the tree according to the sorted order.

5) Create Header Table: Maintain a table containing each frequent item & links to its occurrences in the FP-tree.

Generating FP-tree:-

① Start with the least frequent item in the header table.

② Find its conditional pattern base.

③ Construct FP-tree from the paths.

④ Generate frequent itemsets from the conditional FP-tree.

⑤ Repeat the process for other items until all frequent items are found.

FP-Growth is more efficient than Apriori because it doesn't generate candidate itemsets. It stores the database in a compact FP-tree and requires fewer database scans.

3.b. How are association rules generated from frequent itemsets? Illustrate with suitable example.

Ans: Generation of association rules:-

1) Find frequent itemsets: Use algorithms like Apriori or FP-Growth to find itemsets that satisfy the minimum support.

2) Divide the itemset: Antecedent (X) & consequent (Y): $X \rightarrow Y$

3) Calculate Confidence: $\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$

4) Apply minimum confidence: keeps only rules whose confidence is greater or equal to confidence threshold.

Suppose: $T_1: \{A, B, C\}$, $T_2: \{A, B\}$, $T_3: \{A, C\}$, $T_4: \{A, B, C\}$

minimum support = 50%. Confidence = 60%.

frequent item set $\{A, B\}$

$$A \rightarrow B \Rightarrow 3/3 \times 100 = 100\% \quad (\checkmark)$$

$$B \rightarrow A \Rightarrow 3/3 \times 100 = 100\% \quad (\checkmark)$$

4. Explain the correlation Analysis with suitable examples:

Ans. Correlation analysis is used to find the relationship b/w two variables & determine how strongly they are related.

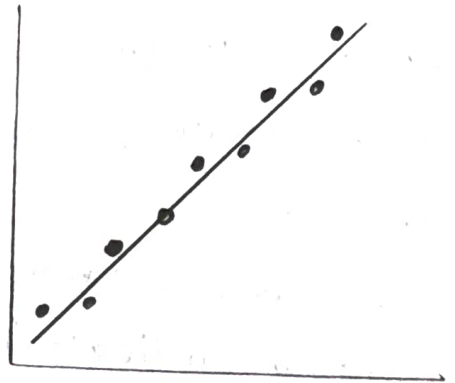
The correlation coefficient is represented by r , and its value lies b/w -1 and +1.

Types of Correlation:

① Positive correlation:

When one variable increases, the other also tends to increase.

Eg. Study time & exam marks. More study time may lead to higher marks.

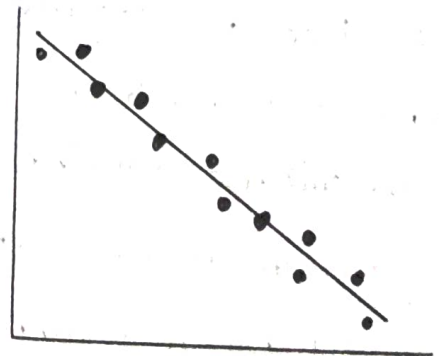


② Negative correlation:

When one variable increases, the other tends to decrease.

Eg. Price of a product, its demand.

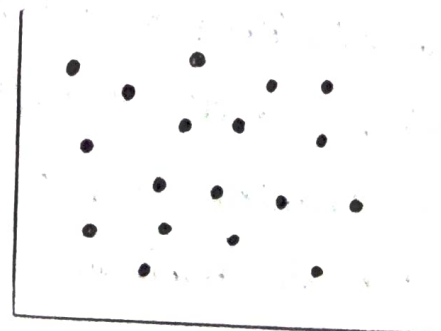
Price $\uparrow$ es, Demand $\downarrow$ es



③ Zero correlation:

There is no clear relationship b/w the two variables.

Eg. A person's shoe size & their exam marks.



Q.
Ans.

Explain the pattern mining in Multilevel Association.

Multilevel association mining discovers relationships b/w items at different levels of abstraction in a concept hierarchy.

For example:

Food $\rightarrow$ Dairy product $\rightarrow$ Milk $\rightarrow$ Low-fat milk

Patterns can be found at a general level or a more specific level.

Steps in Multilevel pattern Mining:

① Create a concept hierarchy: organize items from general to specific levels.

Eg: Food $\rightarrow$ Dairy $\rightarrow$ Milk $\rightarrow$ Low-fat milk

② Find frequent patterns at a higher level:

First, find frequent general items.

Eg: Dairy $\rightarrow$ Bread.

③ Move to lower levels: frequent items are further divided into more specific items.

Eg: Milk $\rightarrow$ Bread

④ Calculate support & confidence

⑤ Generate association rules

Example:

Suppose a supermarket has hierarchy as follows:

Dairy product $\rightarrow$ Milk $\rightarrow$ Low-fat milk

Bakery product $\rightarrow$ Bread $\rightarrow$ Wheat Bread

At a higher level,

Dairy product $\rightarrow$ Bakery products

At lower level,

milk $\rightarrow$ Bread

At an even lower level,

Low-fat milk $\rightarrow$ wheat Bread.